Prabhav Upadhyay

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Research Interests

Experimental aerosol science · Novel atmospheric instrumentation · Aerosol optics · Radiative effects · Aerosol representation in Earth system models · Satellite re-entry · Stratospheric aerosol injection

Education

Washington University in St. Louis 2022 – expected August 2027
PhD in Energy, Environmental & Chemical Engineering; Dean's Select Fellow
Advisor: Prof. Rajan K. Chakrabarty
Dissertation: *Alumina in the upper atmosphere: from proposed stratospheric aerosol injection to emerging satellite re-entry emissions*

Washington University in St. Louis & IIT Bombay 2021 – 2022
M.Tech. in Chemical Engineering, IIT Bombay
M.S. in Aerosol Science and Engineering, Washington University in St. Louis
Advisors: Prof. Rochish Thaokar, Prof. Y. S. Mayya (IIT Bombay); Prof. Jian Wang (WashU)
Thesis: *Development of Acoustic and Electrodynamic Droplet Levitation System for Studies on Droplets and Aerosols*

Indian Institute of Technology (IIT) Bombay 2017 – 2021
Bachelor of Technology in Chemical Engineering; overall CPI: 9.17/10

Research Experience

Washington University in St. Louis Aug 2022 – Present
Graduate Research Assistant, Aerosol Interdisciplinary Research (AIR) Lab
Advisor: Prof. Rajan K. Chakrabarty

Instrumentation

- Redesigned the optical layout with a 1% beam sampler feeding a reference photodiode, enabling a laser-drift correction that lowered the detection limit of a 1047 nm photoacoustic spectrometer from 8.7 to 7.5 Mm^{-1} (≈14% reduction for 1 s measurements) [C2]
- Designed, built, and calibrated integrated photoacoustic nephelometers for simultaneous, real-time measurement of aerosol light absorption and scattering; first-authored a tutorial on photoacoustic spectrometer construction, calibration, and signal processing [J1]
- Developed a COMSOL finite-element model of the photoacoustic resonator, showing that sensitivity is set mainly by the resonator diameter and the acoustic-filter opening [C1]

Aerosol optics for stratospheric aerosol injection (SAI)

- Combined in situ photoacoustic measurements with Mie theory calculations to quantify the weak shortwave absorption of alumina, a proposed SAI material, retrieving imaginary refractive indices of 1.4×10^{-4} to 1.2×10^{-3}; climate model simulations show this absorption has a negligible effect on radiative forcing and stratospheric warming [J3]
- Built Al_2O_3 aggregates by diffusion-limited cluster–cluster aggregation and computed their optics (multi-sphere T-matrix) and radiative forcing (libRadtran), showing that polydisperse monomers raise shortwave forcing by ≈5–25% (median) relative to unaggregated particles, while monodisperse monomers lower it by ≈15–20%, an effect current climate models overlook [C3, C4, M1]
- Developed a framework to quantify the error in radiative forcing from treating polydisperse aerosols as monodisperse [M1]

Satellite re-entry

- Led a study showing that most aluminum particles from satellite re-entry retain a metallic core within an Al_2O_3 shell, rather than being the fully oxidized Al_2O_3 assumed in current climate assessments [C5, M2]

- Integrated plume-oxidation modeling, electron energy-loss spectroscopy of laboratory core–shell analogues, and core–shell Mie theory to show that these particles absorb 30–910 times more sunlight per unit mass than Al_2O_3 [C5, M2]
- Showed, with collaborators' chemistry–climate simulations (SOCOL-AER), that 100 $Gg\,yr^{-1}$ of re-entry particles warms the lower stratosphere by up to 1.6 K, slows the Brewer–Dobson circulation, and redistributes ozone [C5, M2]

Collaborations

- Set up and calibrated the bespoke integrated photoacoustic nephelometer used to measure the optical properties of laboratory-generated, wildfire-like dark brown carbon [J2, S2]
- Co-authored a radiative transfer study showing that purely scattering aerosols can cause positive shortwave forcing over high-albedo surfaces [S1]

Washington University in St. Louis & IIT Bombay Jan 2021 – Feb 2022
Master's Thesis Research
Advisors: Prof. Rochish Thaokar, Prof. Y. S. Mayya (IIT Bombay); Prof. Jian Wang (WashU)

- Developed a non-resonant acoustic levitator that trapped particles in standing sound waves over a wide range of temperature and humidity, using Java simulations of how shifts in transducer frequency and phase change the pressure field
- Fabricated a droplet generator to control the size of charged droplets in an electrodynamic balance, and used MATLAB simulations of gravity's effect on multiple droplets to devise a DC bias on the AC electrode potential that improved imaging of micron-sized droplets

Indian Institute of Technology (IIT) Bombay & Ghent University Apr 2020 – Dec 2020
Undergraduate Research Assistant

- Developed a patient-specific ventilator model to maintain oxygen saturation in COVID-19 patients (Prof. Ravi Gudi, IIT Bombay)
- Devised an algorithm to separate primary from non-primary products in reaction networks (Prof. Joris Thybaut, Ghent University, Belgium)

Publications

Peer-reviewed journal articles

[J3] Kapoor, T. S.*, **Upadhyay, P.***, Huang, J.*, Ren, G., Cavin, J., Mitroo, D., Vattioni, S., Dykema, J., Sedlacek, J., Kumar, J., Hachtel, J. A., Xu, L., Mishra, R., & Chakrabarty, R. K. (2026). Unveiling the shortwave absorption spectra of alumina aerosols: Radiative implications for solar radiation modification. *Geophysical Research Letters*, *53*(16), e2026GL121668. doi:10.1029/2026GL121668. *Equal contribution. ***Co-first author***

[J2] You, S.-W., **Upadhyay, P.**, Kapoor, T. S., Cheng, Z., Ren, G., Puthussery, J. V., Sumlin, B. J., China, S., Mishra, R., & Chakrabarty, R. K. (2025). Laboratory synthesis and characterization of wildfire-like dark brown carbon aerosols. *Aerosol Science and Technology*, *59*(12), 1553–1565. doi:10.1080/02786826.2025.2534587 ***Co-author***

[J1] **Upadhyay, P.**, Sumlin, B. J., Kapoor, T. S., Yang, L., & Chakrabarty, R. K. (2025). Tutorial: A photoacoustic spectrometer to measure aerosol light absorption. *Journal of Aerosol Science*, *185*, 106518. doi:10.1016/j.jaerosci.2024.106518 ***First author***

Submitted manuscripts and preprints

[S2] You, S.-W., Kapoor, T. S., **Upadhyay, P.**, Chou, L.-T., Cheng, Z., Zelenyuk, A., Ijaz, A., Vandergrift, G., Kulkarni, G., Niu, S., Hsiao, T.-C., China, S., & Chakrabarty, R. K. (2026). Influence of photochemical aging on light absorption and cloud activation properties of dark brown carbon. Submitted to *Environmental Science & Technology*.

[S1] Zhang, L., Kapoor, T. S., **Upadhyay, P.**, & Chakrabarty, R. K. (2026). Purely light-scattering aerosols could induce positive shortwave radiative forcing over high-albedo surfaces. *EGUsphere* [preprint], in review for *Atmospheric Chemistry and Physics*. doi:10.5194/egusphere-2026-3947

Manuscripts in preparation

[M2] **Upadhyay, P.**, et al. Enhanced stratospheric warming driven by satellite re-entry composite aerosols. In preparation. Summary online

[M1] **Upadhyay, P.**, et al. Monomer polydispersity enhances the shortwave radiative efficiency of fractal aggregates for stratospheric aerosol injection. In preparation. Summary online

Conference Presentations

[C5] **Upadhyay, P.**, Huang, J., Vattioni, S., Su, W., You, S.-W., Sukhodolov, T., Dykema, J., Sedlacek, J., Seliner, K., Kapoor, T. S., Liang, X., Mishra, R., & Chakrabarty, R. K. (2026, October). *Metal rain: How satellite re-entry could warm the stratosphere* [Poster presentation]. 44th Annual Conference of the American Association for Aerosol Research (AAAR), Pasadena, CA.

[C4] **Upadhyay, P.**, Kapoor, T. S., & Chakrabarty, R. K. (2026, April). *Polydisperse monomers improve shortwave radiative efficiency* [Poster presentation]. Simons Foundation SRM Annual Meeting, New York, NY.

[C3] **Upadhyay, P.**, Kapoor, T. S., & Chakrabarty, R. K. (2025, December). *Could aggregation and sulfate coating alter the scattering and absorption efficiencies of SAI aerosol?* [Oral presentation]. AGU Fall Meeting, New Orleans, LA.

[C2] **Upadhyay, P.**, Kapoor, T. S., & Chakrabarty, R. K. (2024, October). *Enhancing the measurement accuracy of integrated photoacoustic nephelometers to better characterize infrared optical properties of stratospheric aerosol injection candidates* [Poster presentation]. 42nd AAAR Annual Conference, Albuquerque, NM.

[C1] **Upadhyay, P.**, Puthussery, J. V., Kapoor, T. S., Sumlin, B. J., & Chakrabarty, R. K. (2023, October). *A computational sensitivity study to refine photoacoustic absorption signals through geometry optimization* [Oral presentation]. 41st AAAR Annual Conference, Portland, OR.

Teaching Experience

Washington University in St. Louis

Teaching Assistant, Energy, Environmental & Chemical Engineering

- Transport Phenomena II: Energy and Mass Transfer — Fall 2024
- Transport Phenomena I: Basics and Fluid Mechanics — Spring 2024

Held office hours, graded homework and exams, and assisted with in-class experiments.

Lab mentoring: trained two junior researchers in experimental aerosol methods, instrument operation, and maintenance

Indian Institute of Technology (IIT) Bombay

Teaching Assistant

- Introduction to Data Analysis (Chemical Engineering) — Spring 2022
  Held office hours; graded homework and exams
- Differential Equations II (Mathematics) — Fall 2019
  Helped design exams and evaluate quizzes and finals; led tutorials for a class of 100 students

Service

Judge, poster and oral presentations — Oct 2026
44th AAAR Annual Conference, Pasadena, CA

Technical Skills

Instrumentation & optics: photoacoustic spectrometry · integrated photoacoustic nephelometers · acoustic resonators · acoustic levitator · electrodynamic balance · laser & photodetector integration · sensor circuit design · optical alignment · calibration & detection limits (Allan deviation)

Aerosol generation & sizing: dry-powder dispersion · SMPS · AAC · cyclones & charge neutralizers · TEM grid sampling

Modeling & simulation: COMSOL Multiphysics (acoustics, CFD) · Mie theory & inversion (PyMieScatt) · multi-sphere T-matrix · discrete dipole approximation · aerosol dynamics (deterministic & Monte Carlo) · chemistry box model · libRadtran / DISORT radiative transfer

Programming & hardware: C++ · LabVIEW · Python · MATLAB · Arduino · DAQ-based instrument automation